MOVES6 Database Conversion Tools Help

Introduction

The database conversion tools supplied with MOVES6 can be used to convert input databases developed with any version of MOVES4 or MOVES5 to be compatible with MOVES6. However, these tools should only be used when the input databases developed for MOVES4 or MOVES5 still contain the latest local information. If you have newer data, EPA recommends that you create a new input database using the MOVES6 interface.

Using the Tools

The tools are accessible via the Tools menu in the MOVES Graphical User Interface (GUI):

- “Convert MOVES4 Input Database to MOVES6”
- “Convert MOVES5 Input Database to MOVES6”

Select the appropriate tool based on what version of MOVES was used to develop the old input database. Then follow the instructions provided in the GUI for selecting the database to be converted, entering the name of the new database to be created, and running the tool.

Additional Steps Necessary to Use Converted Databases with MOVES6

After running the tool, additional work is required to use the converted databases with MOVES6. First, you will need to select the newly converted database on the Create Input Database panel.

> **Note**: If this panel is grayed out, you will need to completely fill out the RunSpec in accordance with the data in your converted database so that you get green checks for all the other panels.

Once on the Create Input Database Panel, you may need to click the Refresh button if your new database does not automatically appear in the list. After selecting it, click the Enter/Edit Data button to open the County, Project, or Nonroad Data Manager (depending on the type of RunSpec being edited). The following sections detail the tabs that may need additional attention before running the model. See the MOVES6 Technical Guidance: Using MOVES to Prepare Emission Inventories for State Implementation Plans and Transportation Conformity (EPA-403-B-26-002) for more information.

Note that many of the following sections discuss age groups and/or model years. This is because MOVES4 and earlier versions only accounted for age groups 0-30 and the oldest model year that they covered was 1960. MOVES5 and later versions account for age groups 0-40 and the oldest model year it covers is 1950. This is relevant when converting tables from MOVES4 to MOVES6 that contain data by age or model year.

Fuels

- The fuels tables are not converted by the tool; instead, it simply produces empty fuels tables. From the Fuels Tab, export the default fuels, review them, and make any changes as necessary to the *AVFT*, *FuelFormulation*, *FuelSupply*, and *FuelUsageFraction* tables, following the technical guidance.

    - The AVFT Tool can be used to incorporate the latest fuel type distributions and projections.
- When ready, import the fuels data.
    - After importing, use the Fuels Wizard to make any changes to the fuel formulation parameters if necessary.

Age Distribution

- In general, the data in your input database's *SourceTypeAgeDistribution* table are carried over to the new database. However, the tool makes the following changes:
    - Following the MOVES6 Technical Guidance, the tool uses the MOVES6 national default age distributions for the long-haul source types (53 and 62) instead of the data in your input database.
    - Age distributions prepared for MOVES4 accounted for ages 0-30, and the fraction assigned to age 30 included all vehicles ages 30+. MOVES6 requires age distributions to include ages 0-40, with the fraction assigned to age 40 to include all vehicles ages 40+. Therefore, when converting MOVES4 input databases, the tool estimates age fractions in the new database for ages 30-40 based on the age 30 fractions in your input database. The specific algorithm used for this estimation is described in Population and Activity of Onroad Vehicles in MOVES6 (EPA-420-R-26-010).
- **Important**: If you have the original data used to derive the 0-30 age distributions, you will get more accurate results if you reanalyze the data following the MOVES6 Technical Guidance to develop 0-40 age distributions. To use your reanalyzed data, click the Clear Imported Data button for the *SourceTypeAgeDistribution* table and then import your reanalyzed data.

I/M Programs

- The data in your input database's *IMCoverage* table are carried over to the new database.
- For MOVES4 input databases, if any I/M programs are applicable to model year 1960, the transferred data are updated to apply to model year 1950.
- **Important**: If the inputs in your MOVES4 table were based on previous model defaults, these data should be discarded and the MOVES6 defaults should be used instead. To do so, click the Clear Imported Data button for the IMCoverage table, export the default data, review and make any necessary changes, and then import this table. Note that there were no changes to the default IMCoverage data between MOVES5 and MOVES6.

Hotelling

Data for any of the optional hotelling tables in MOVES5 input databases are carried over to the new database as-is. The sections below detail additional changes when converting MOVES4 input databases.

HotellingActivityDistribution

- If this optional table contains local data that go back to model year 1960, the transferred data are updated to cover model year 1950. No additional steps are necessary when converting this table.

HotellingAgeFraction

- The *HotellingAgeFraction* is an optional table. If it is provided, MOVES6 requires it to contain data for ages 0-40.
- MOVES4 only covered ages 0-30 and this tool is not able to estimate activity for ages 31-40 for this table. Therefore, if you wish to use this table with MOVES6, you will need to reanalyze your local hotelling data to provide data for all ages 0-40, and then import the reanalyzed data using the MOVES6 data importer.

Idle

IdleModelYearGrouping and TotalIdleFraction

- The *IdleModelYearGrouping* and *TotalIdleFraction* are optional tables. If either are provided, the MOVES5 converter tool will carry over the data as-is to the new database.
- When using the MOVES4 converter tool, if either of these tables contain local data that cover model year 1960, the transferred data are updated to cover model year 1950. No additional steps are necessary when converting these tables.

Starts

Starts and StartsAgeAdjustment

- The *Starts* and *StartsAgeAdjustment* are optional tables. If either are provided, MOVES6 requires them to contain data for ages 0-40. The MOVES5 converter tool will carry over the data as-is to the new database.
- MOVES4 only covered ages 0-30 and this tool is not able to estimate activity for ages 31-40 for these tables. Therefore, if you wish to use either of these tables with MOVES6, you will need to reanalyze your local starts data to provide data for all ages 0-40, and then import the reanalyzed data using the MOVES6 data importer.

StartsOpModeDistribution

- The *StartsOpModeDistribution* is an optional table. If it is provided, MOVES6 requires it to contain data for ages 0-40.
- MOVES4 only covered ages 0-30. Therefore, when converting MOVES4 input databases, the tool will create soak distributions for ages 31-40 using the age 30 soak distributions in your input data. No additional steps are necessary when converting this table.